

Roll No. 11 (a): Basic Data Pre-Processing Using NumPy

Aim

To perform basic data pre-processing operations using the NumPy library.

Program

```
import numpy as np

# Create array
data = np.array([10, 20, 30, 40, 50])

print("Original Data:")
print(data)

print("Mean =", np.mean(data))
print("Median =", np.median(data))
print("Maximum =", np.max(data))
print("Minimum =", np.min(data))

print("\nReshaped Array:")
print(data.reshape(5,1))
```

Output

```
Original Data:
[10 20 30 40 50]

Mean = 30.0
Median = 30.0
Maximum = 50
Minimum = 10
```

Reshaped Array:

```
[[10]  
 [20]  
 [30]  
 [40]  
 [50]]
```

Conclusion

Thus, basic data pre-processing operations using NumPy were performed successfully.

Viva Questions

1. What is NumPy?

NumPy is a Python library used for numerical computations.

2. What is an array?

An array is a collection of similar data elements.

3. What does `np.mean()` do?

It calculates the average value.

4. What does `np.median()` do?

It returns the middle value of the dataset.

5. What does `np.max()` do?

It returns the largest value.

6. Why is `reshape()` used?

To change the dimensions of an array.

Roll No. 11 (b): Support Vector Machine (SVM)

Aim

To build a Support Vector Machine (SVM) classifier and classify a new instance.

Program

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC

# Load dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create SVM model
model = SVC(kernel='linear')

# Train model
model.fit(X_train, y_train)

# New instance
new_instance = [[5.1, 3.5, 1.4, 0.2]]

# Predict
prediction = model.predict(new_instance)

print("Predicted Class:", iris.target_names[prediction[0]])
```

Output

```
Predicted Class: setosa
```

Conclusion

Thus, the Support Vector Machine classifier was implemented successfully and used to classify a new instance.

Viva Questions

1. What is SVM?

SVM is a supervised learning algorithm used for classification.

2. What is a hyperplane?

A boundary that separates different classes.

3. What is a kernel?

A function used to separate complex data.

4. Which kernel is used in this program?

Linear kernel.

5. Is SVM supervised or unsupervised?

It is a supervised learning algorithm.

6. What is classification?

The process of assigning data to predefined classes.

7. What is `model.fit()`?

It trains the model using training data.

8. What is `model.predict()`?

It predicts the class of new data.